

ALI RAZA CHAUDHRY

AI/ML Engineer | Data Scientist | Computer Vision | Generative AI | Full Stack Developer

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PROFESSIONAL SUMMARY

AI/ML Engineer and Data Scientist with hands-on experience in machine learning, deep learning, computer vision, data analytics, statistical modeling, and generative AI. Experienced in developing end-to-end solutions spanning data preparation, exploratory and statistical analysis, model development, evaluation, optimization, and deployment. Proficient in Python, PyTorch, TensorFlow, OpenCV, YOLO, R, SQL, LLMs, RAG, and AI agents, with additional full-stack expertise in FastAPI, React, Next.js, and Node.js. Focused on building scalable, data-driven, and production-ready AI systems with measurable performance and practical business impact.

EXPERIENCE

Freelance Computer Vision Engineer | Self-Employed / Fiverr

2025 – Present

- Built YOLO-based computer vision solutions with FastAPI deployment.
- Built a golf swing tracking system using pose estimation to analyze swing angles and tempo for coaching feedback.

Embedded Systems Intern | Meta Desk Global — Lahore, Pakistan

July 2025 – July 2026

- Worked on AI/ML, Data Science, Computer Vision, and NLP tasks involving data preprocessing, model development, evaluation, and optimization.
- Contributed to microcontroller programming, circuit interfacing, and hardware debugging as part of embedded systems development.

PROJECTS & PORTFOLIO

UFC Vision AI — Real-Time Fight Analytics

- Custom YOLOv11 fighter-tracking and action-detection (punches, kicks, grappling) with pose estimation, fight stats, and a React dashboard.

AI Cricket Match Analytics System

- YOLOv8 + OpenCV pipeline for player/ball detection, field mapping, shot analysis, pitch maps, and automated highlight generation.

Smart Football Match Analysis System

- Broadcast-footage analytics: player/ball tracking, team classification, bird's-eye-view transform, speed estimation, heatmaps, event detection.

AI Skeleton Detection & 3D Avatar Generation

- Real-time MediaPipe Pose pipeline (30+ FPS), 3D avatar generation with Trimesh/Open3D, coach-correction blending, FastAPI integration.

Smart Vehicle Gate Access System (Final Year Project)

- License-plate detection and vehicle make/model recognition (VMMR) for automated, all-weather gate access. | automated-gate-access.vercel.app

AI Baseball Swing Performance Analyzer

- CV pipeline for batting video: pose estimation, swing-phase and strike-zone detection, bat/ball tracking, biomechanics scoring.
- MediaPipe Pose, YOLO, and GPU-accelerated inference with annotated videos and a Streamlit dashboard.

Additional Computer Vision Projects

- **Automatic Product Scanning & Billing System** — vision-based retail checkout automation.
- **Real-Time Multi-Person Tracking System** — multi-object tracking with analytics output.
- **Automatic Vehicle Counting System** — real-time traffic counting and analytics.
- **Vehicle Make and Model Classification** — fine-grained vehicle image classification.
- **Vehicle Damage Detection and Cost Estimator** — damage localization with automated cost estimation.
- **Real-Time Tools & People Detection** — object detection for safety and analytics.

- **Waste Classification & Furniture Classification** — visual sorting and category recognition.
- **Bayesian Statistical Analysis & Predictive Modeling** — posterior estimation and hypothesis testing on real-world data.

Full-Stack Web Applications

- **Crown and Dial** — Luxury e-commerce storefront (React). | crown-and-dial.netlify.app
- **Fine Crafted Structures** — Luxury construction company site (Next.js). | constructioncompany.netlify.app
- **Healthy Start** — Non-Emergency Medical Transportation platform (WordPress/PHP). | healthystartnc.com
- **Nyuton Enterprises** — Build Real Cashflow (WordPress/PHP). | nyutonenterprises.com
- **Fine Crafted Structures** — Full-service construction firm (React). | finecraftedstructures.netlify.app
- **Stepping Stones CRI** — Mental Health, Well-being & Prevention (React). | steppingstone.netlify.app
- **Stepping Stones CRI** — Premium website (React). | steppingstoness.netlify.app
- **Revara** — Real estate investment platform (Next.js). | hotel-web-chi-tan.vercel.app
- **NFT Marketplace** — Discover, create & trade extraordinary NFTs (React.js). | nftmarketplaece.netlify.app
- **Techlo** — Cutting-edge online solutions agency (React.js). | techloagenccy.netlify.app
- **Clipzy** — AI-powered live streaming & content studio (Next.js, React, Node.js, AI APIs). | clipzy.xynovix.com
- **FamLink** — Family care platform connecting families with caregivers. | famlink.care
- **NOVA — Premium Footwear** — Interactive 3D web experience with real-time rendering. | core-seven-henna.vercel.app
- **Luxury Watch Store** — Modern e-commerce storefront (Next.js, React). | watchs-gray.vercel.app

TECHNICAL SKILLS

Computer Vision: OpenCV, MediaPipe, YOLOv8/v11, RF-DETR, Object Detection, Image Classification, Image Segmentation, Multi-Object Tracking, OCR, Real-Time Vision Pipelines

Machine Learning: Supervised & Unsupervised Learning, Regression, Classification, Clustering, Decision Trees, Random Forest, XGBoost, LightGBM, SVM, KNN, Naive Bayes, Ensemble Learning, Dimensionality Reduction, PCA, Feature Selection, Model Evaluation & Optimization

Deep Learning: PyTorch, TensorFlow, Keras, CNNs, RNNs, LSTMs, GRUs, Autoencoders, GANs, Transfer Learning, Attention Mechanisms, Transformers, Neural Networks

Generative AI & LLMs: Generative AI, Large Language Models (LLMs), Prompt Engineering, RAG, Vector Databases, Embeddings, Transformers, AI Agents, Agentic AI, LangChain, LangGraph, Hugging Face, Fine-Tuning, LLM APIs

Data Science & Analytics: Data Analysis, Data Analytics, EDA, Data Cleaning, Data Preprocessing, Feature Engineering, Predictive Analytics, Data Visualization, Experiment Tracking

Statistical Analysis: Descriptive & Inferential Statistics, Hypothesis Testing, t-tests, ANOVA, Chi-Square Tests, Correlation & Regression, Probability, Confidence Intervals, Bayesian Analysis, A/B Testing

Statistical & Analytics Tools: R, RStudio, Stata, Python, Jupyter, Power BI, Tableau, Excel

Python Data Libraries: NumPy, Pandas, SciPy, Statsmodels, Scikit-learn, Matplotlib, Seaborn, Plotly

Languages & Backend: Python 3.9+, FastAPI, JavaScript (ES6+), Node.js, Express.js

Frontend: React, Next.js, HTML, CSS, Tailwind, Bootstrap

Databases: SQL, PostgreSQL, SQL Server, MySQL, MongoDB, NoSQL

Tools: Git, GitHub, VS Code, Postman, REST APIs, Webhooks

Cloud & SaaS: AWS, SaaS Platforms, Docker, Kubernetes, CI/CD, Linux

EDUCATION

Bachelor of Computer Engineering | COMSATS University, Lahore | CGPA: 2.81/4.00

June 2026

CERTIFICATIONS

- Python Machine Learning — Udemey
- OpenCV Bootcamp — OpenCV University
- Artificial Intelligence — UniAthena